2026-10-06 | VS&A Daily

Random vibration test band splitting



GPM Core Observatory on a horizontal vibration shaker at NASA Goddard, July 2013. The photograph shows the scale of the hardware and fixture involved; it is not evidence that this test used band splitting. Credit: NASA/Warren Shultzaberger. Image source [5].

The shaker gets a break. Does the hardware?

A vibration specification may fit comfortably on one sheet of paper and still ask more of a shaker than it can deliver. Dividing the spectrum into successive runs can make the test feasible. The interesting question is what the hardware loses when frequencies that once arrived together take turns.

This bonus lesson follows that trade: how to separate a spectrum without quietly separating the loads that matter. By the end, you should be able to challenge a proposed split using response bandwidth, exposure time and the failure mechanism being verified.

Start with the reason for splitting

Here, band splitting means applying different portions of a random vibration spectrum in successive runs along the same axis. It is distinct from splitting a spectrum into frequency bins for analysis, or driving several bands simultaneously. The distinction sounds small until you ask a joint, circuit board or mechanism to respond.

The benefit is lower simultaneous demand. But the limiting quantity might be force, displacement, velocity, amplifier current or fixture response. A lower acceleration RMS alone does not demonstrate enough facility margin. The feasibility assessment has to follow the actual limit through the combined shaker, fixture and article.

The spectrum adds. The test does not.

For a linear, time-invariant structure under stationary random excitation, a scalar stress response is obtained by weighting the input acceleration PSD with the squared magnitude of its stress transfer function. That gives us a clean starting point:

$$\sigma_{\text{rms}}^2 = \int_0^\infty |H_{\sigma a}(f)|^2 G_a(f)\, df$$

Here f is frequency in Hz, G_a is a one-sided acceleration PSD in g^2/Hz, and $H_{\sigma a}$ is stress per unit base acceleration in MPa/g. The integral gives stress variance in MPa^2. For ideal complementary bands, their separate variances add to the full-band variance. Their RMS values do not add directly.

That identity is an analysis result, not proof of test equivalence. A full-band run produces the sum of the band responses at every instant. Sequential runs never produce that simultaneous sum. They can reproduce each frequency's exposure while changing stress amplitudes, extremes and cycles.

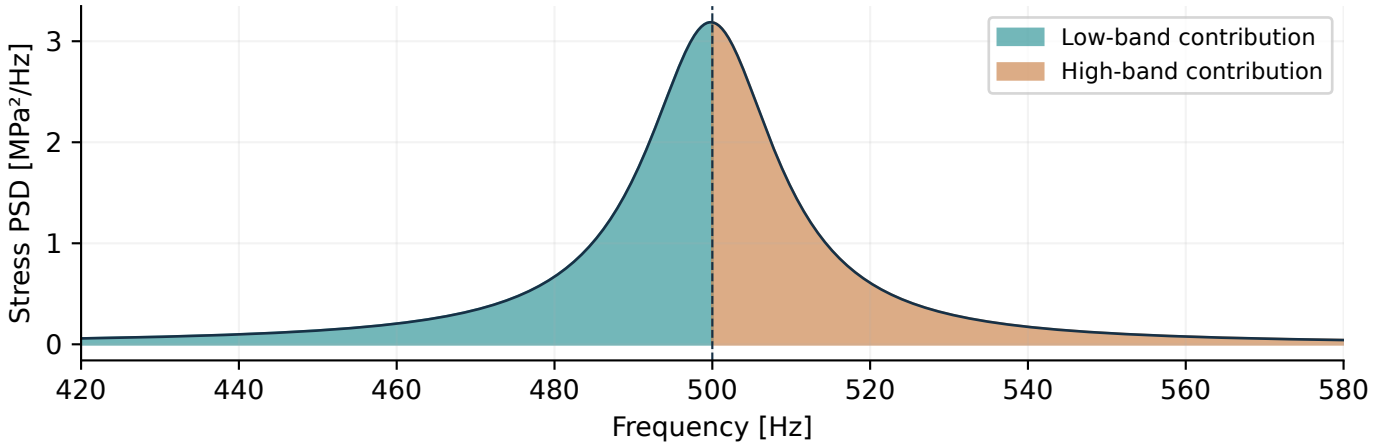


Figure 1. Original illustrative modal stress PSD: a 500 Hz mode with 2% damping, excited by a flat 20-2,000 Hz input and split at 500 Hz. Each run supplies only part of the same resonance. The response is normalized to 10 MPa RMS under full-band excitation. No measured data.

A lightly damped mode is a frequency filter, but it is not infinitely narrow. Cutting near its natural frequency divides the response that drives its local stress. Splitting between well-separated response peaks is more promising: each important mode can then receive nearly its full excitation in one run. The relevant peaks may be local component modes, not just the payload's primary modes.

Two minutes can be less severe than one

Take an illustrative flat input of 0.04 g²/Hz from 20 to 2,000 Hz for 60 seconds. Its overall level is 8.90 g RMS. A sharp split at 500 Hz produces 4.38 g RMS below the split and 7.75 g RMS above it. Run each band for the original 60 seconds: every frequency gets its original exposure, and total running time becomes 120 seconds.

Now use the mode in Figure 1. Its transfer-function shape is the reciprocal of the standard damped-oscillator dynamic stiffness; the stress scale is chosen only to make the full response 10 MPa RMS. Numerical integration gives about 52.9% of the stress variance below the split and 47.1% above it. The separate responses are 7.27 and 6.86 MPa RMS at this stress location.

For narrowband Gaussian stress, a power-law stress-life model and linear damage accumulation give the following scaling. The exponent b describes fatigue sensitivity; ν is the expected cycle rate and T is duration. Material constants cancel in a comparison using the same model.

$$D \propto T \nu \sigma_{\text{rms}}^{b}$$

To see the mechanism without numerical clutter, let both bands contribute half the stress variance and have approximately the same cycle rate. With b = 6 and each band run for T, the resulting damage ratio is:

$$\frac{D_{\text{split}}}{D_{\text{full}}} \approx 2\left(\frac{1}{\sqrt{2}}\right)^{6} = \frac{1}{4}$$

Twice the elapsed time, roughly one-quarter of the estimated fatigue damage. Keeping the small differences in variance and spectral zero-crossing rate in this example gives approximately 0.25 as well. This is a narrowband fatigue estimate, not a measured life prediction or an exact rainflow result. Irvine provides a separate, Dirlik-based band-splitting study for comparison [3].

The useful lesson is not to multiply every split test’s duration by four. A mode well inside one band may already receive almost all its original fatigue exposure. Extending both runs indiscriminately could over-test that mode while still failing to reproduce simultaneous peak loads or functional interactions elsewhere.

Move the split, and the answer changes

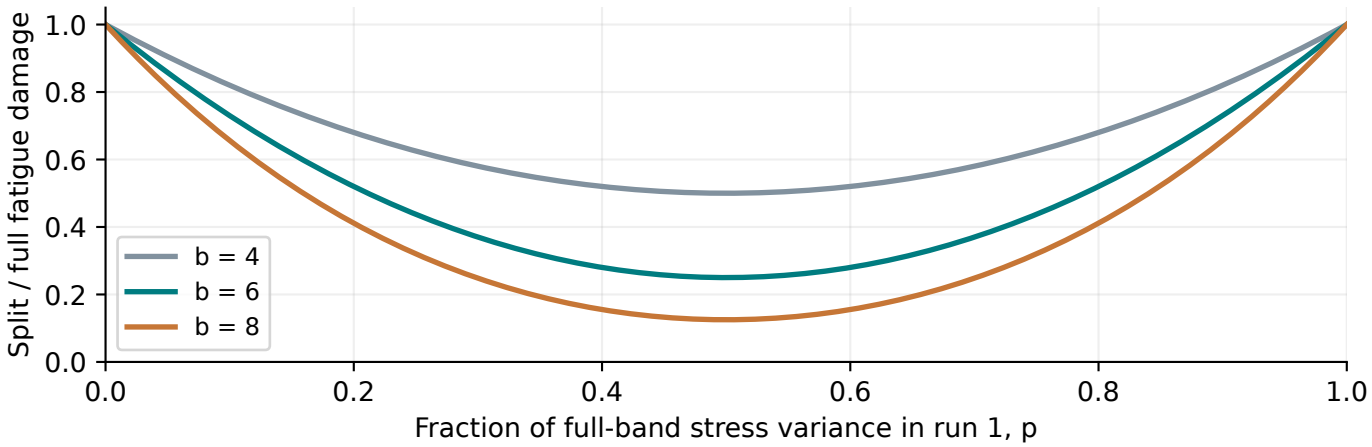


Figure 2. Original analytical illustration: $p^{b/2} + (1-p)^{b/2}$, for equal durations and approximately equal cycle rates. The greatest deficit occurs when a split divides a stress response equally. This curve is a mechanism demonstration, not a universal correction factor.

Overlap can protect a resonance from a hard edge, but it also repeats exposure. If both runs apply full PSD in their overlap for the original duration, that region receives twice the spectral exposure. A background floor likewise adds exposure outside the active band. Include both in response and fatigue calculations; do not simply join the largest portions of two plotted spectra and call the test recovered.

The published rules are deliberately specific

SpaceX's Falcon guide, section 7.4.5.2 [1], allows case-by-case splitting for integrated random vibration tests when shaker force capability requires lower interface loads. Its constraints include no more than three bands, at least one-third-octave overlap, avoiding payload primary modes, and using the same article. Every run retains a 0.004 g^2/Hz workmanship floor across 20-2,000 Hz; qualification/protoqualification should target the maximum predicted environment (MPE) overall RMS. These are Falcon verification provisions, not general rules for all hardware.

NASA Marshall's MSFC-STD-3676B [2] discourages splitting but recognizes that it may be necessary. It calls for reviewing unnecessary conservatism, axis-specific criteria and fixture mass; using measured resonances to place breaks; and obtaining approval from the responsible organization. Its scope is MSFC-managed launch vehicle and payload hardware and contracts that levy it. Neither document makes spectral bookkeeping a substitute for response assessment.

For perspective, a one-third-octave interval has an upper-to-lower frequency ratio of about 1.26. An overlap geometrically centered on 500 Hz would span roughly 445-561 Hz. That illustrates the width only: centering a split on a primary mode would conflict with the Falcon provision, and modal uncertainty still needs consideration.

Keep the failure mechanism in view

There is a second, very useful role for splitting: diagnosis. Smith and colleagues reported using separate bands while investigating a heritage thrust-vector-control actuator for SLS [4]. Isolating a roughly 250 Hz feedback-mechanism resonance helped establish that resonance was not the only problem. Low-frequency inertial loads overcoming mechanism preload were the dominant source of functional anomalies. A diagnostic split answered a causal question; it did not establish general qualification equivalence.

For a proposed qualification split, I would want one coherent comparison: original versus proposed response PSDs at critical locations, accumulated duration and fatigue exposure, and the simultaneous-load mechanisms that the substitute cannot exercise. Include uncertain modal frequencies and damping, realistic controller transitions, overlap, background floors and functional monitoring. Low-level surveys help position the bands, but amplitude-dependent joints or contact behavior can move the problem once the level rises.

Three ideas to carry into the morning

1. Keep each active band's intended exposure; dividing the original time among bands is an additional reduction.
2. Place breaks using response bandwidth and uncertainty, then account for repeated overlap exposure.
3. Judge equivalence against the failure mechanism. Fatigue, peak load and functional performance need different evidence.

Reflection: Both band runs meet their PSD tolerances. What remains unproven?
Answer: That critical responses, cumulative damage and simultaneous-load behavior adequately represent the full-band test. Input compliance alone cannot answer those questions.

Sources and calculation notes

[1] SpaceX, Falcon User's Guide, file dated May 9, 2025, §7.4.5.2, printed p. 56. Mission-specific applicability governs.

[2] NASA/MSFC, MSFC-STD-3676B, March 15, 2017, §§1 and 7.2, pp. 8, 32-33. Cited revision; verify the version levied by your program.

[3] Tom Irvine, PSD Band-Splitting Example, November 11, 2014. Original response/fatigue study; its numerical example and historical band limits are not adopted here.

[4] A. Smith et al., Significant Attenuation of Lightly Damped Resonances Using Particle Dampers, 29th Aerospace Testing Seminar, October 2015, pp. 3, 9, 22.

[5] NASA, GPM Completes Vibration Tests, July 26, 2013. Opening photograph: NASA/Warren Shultzaberger.

Original calculations assume a linear stationary Gaussian response, ideal complementary bands, 2% damping and an illustrative b = 6. No mean-stress, nonlinear contact or multiaxial fatigue model is included. Sources accessed October 6, 2026. GPT-6 Astra selection could not be verified for this bonus edition.